Experimental Verification of Metatime Topological Eigenvalues via Optical Solitons in SSH Waveguide Lattices

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Abstract

We propose a direct experimental test of the Metatime framework by mapping topological eigenvalues τ_i to winding numbers N_{wind} in Su-Schrieffer-Heeger (SSH) photonic waveguide lattices. By tuning SSH lattice parameters to match Metatime predictions and measuring soliton dynamics, we can verify the topological protection mechanism underlying fermion mass generation. This approach provides experimental validation independent of DUNE neutrino measurements, with timeline of 6–12 months.

1 Introduction

The Metatime framework predicts that Standard Model fermion masses emerge from topologically quantized eigenvalues on a Kähler manifold:

$$\tau_i \in \{4, 1, 10\} \quad \text{(electrons, muons, taus)} \tag{1}$$

These eigenvalues are *topologically locked*—perturbations violate topological constraints, causing systematic collapse of predictions. This suggests an underlying topological protection mechanism amenable to direct experimental observation in optical systems.

2 Physical Principle

2.1 Mapping Metatime to SSH Solitons

The SSH model describes tight-binding electrons with alternating hopping strengths in a 1D chain:

$$H_{\text{SSH}} = \sum_n \left[t_A c_n^\dagger c_{n+1} + t_B c_{n+1}^\dagger c_n \right] \tag{2}$$

where t_A, t_B are intradimer and interdimer couplings. The topological phase exhibits boundary states with quantized properties.

For optical implementation, photonic SSH lattices are fabricated in waveguide arrays, where:

- Waveguide separation $\leftrightarrow$ coupling strength t_A, t_B
- Photon propagation $\leftrightarrow$ electron dynamics
- Winding number $N_{\text{wind}} \in \mathbb{Z} \leftrightarrow$ topological invariant

2.2 Connection to Metatime Eigenvalues

We propose identifying:

$$\tau_e \leftrightarrow N_{\text{wind},e} = 4 \tag{3}$$

$$\tau_\mu \leftrightarrow N_{\text{wind},\mu} = 1 \tag{4}$$

$$\tau_\tau \leftrightarrow N_{\text{wind},\tau} = 10 \tag{5}$$

If this mapping is valid, optical solitons in SSH waveguides tuned to Metatime parameters should exhibit:

1. Quantized winding numbers matching τ_i
2. Topological protection under disorder and rapid variations
3. Systematic collapse when parameters deviate from τ_i

3 Experimental Protocol

3.1 Phase 1: SSH Waveguide Fabrication (Weeks 1–8)

Specification:

- Material: Nd:$BaLaGa_3O_7$ crystal (solid-state, room temperature)
- Technique: Femtosecond laser writing (established method [2])
- Waveguide count: 60 channels (3 SSH domains for $\tau_e, \tau_\mu, \tau_\tau$)
- Lattice constant: $\Lambda = 20\,\mu$m (optimized for 1550 nm operation)

Parameter tuning:
For each fermionic flavor $i \in \{e, \mu, \tau\}$, configure SSH parameters to produce winding number $N_{\text{wind},i} = \tau_i$:

$$t_A^{(i)} = \frac{\hbar\omega_0}{N_{\text{wind},i}} \cdot f_A(\tau_i) \tag{6}$$

$$t_B^{(i)} = \frac{\hbar\omega_0}{N_{\text{wind},i}} \cdot f_B(\tau_i) \tag{7}$$

where f_A, f_B are functions determined from SSH topological condition:

$$\frac{t_B}{t_A} \gtrless 1 \quad \Leftrightarrow \quad \text{topological/trivial phase} \tag{8}$$

3.2 Phase 2: Temporal Soliton Excitation (Weeks 9–16)

Setup:

- Pump laser: 1550 nm, 45.3 W peak power (per recent experiments [1])
- Pulse width: 0.93 ps input (compressed to ~ 0.47 ps in topological mode)
- Nonlinearity: Kerr effect in crystal ($n_2 \sim 10^{-19}\,\text{m}^2/\text{W}$)

Protocol:
For each SSH domain i:

1. Inject 1550 nm pulse at boundary waveguide (topological edge mode)

2. Vary pump power P from 10 W to 50 W in steps of 5 W
3. Measure output temporal compression factor
4. Record spatial profile of soliton localization

Expected signal:
Soliton formation occurs when:

$$\text{GVD} + \text{Kerr dispersion} + \text{Topological confinement} = 0 \tag{9}$$

At critical power $P_c^{(i)}$, soliton profile $u(t, z)$ satisfies:

$$\frac{\partial u}{\partial z} + \frac{i\beta_2}{2}\frac{\partial^2 u}{\partial t^2} + i\gamma|u|^2 u = 0 \tag{10}$$

with boundary condition enforcing $N_{\text{wind}} = \tau_i$.

3.3 Phase 3: Winding Number Measurement (Weeks 17–24)

Technique: Spatial Light Modulator (SLM) Phase Reconstruction
Extract winding number by measuring phase gradient around topological mode:

$$N_{\text{wind}} = \frac{1}{2\pi}\oint_C \nabla \arg\{u(x, y)\} \cdot d\ell \tag{11}$$

where C is closed contour encircling topological edge mode.
Procedure:

1. Use near-field scanning optical microscopy (NSOM) to map amplitude $|u(x, y)|$
2. Use phase-shifting interferometry to map phase $\arg\{u(x, y)\}$
3. Compute line integral numerically
4. Repeat for each power level and each SSH domain

Predicted results:

Domain	τ_i (Metatime)	N_{wind} (measured)	Tolerance
e	4	4.0 ± 0.1	$\pm 2.5\%$
μ	1	1.0 ± 0.1	$\pm 10\%$
τ	10	10.0 ± 0.2	$\pm 2\%$

Table 1: Expected winding number measurements matching Metatime eigenvalues

3.4 Phase 4: Robustness Tests (Weeks 25–32)

Test A: Anderson Disorder
Introduce controlled disorder in intradimer couplings (Bernoulli distribution):

$$t_A^{(i)} \to t_A^{(i)}(1 + \delta_n), \quad \delta_n \in \{-\epsilon, +\epsilon\} \tag{12}$$

Vary disorder strength ϵ from 0 to 30%. Prediction: - N_{wind} remains locked to τ_i for $\epsilon < 20\%$ - Topological protection breaks sharply at $\epsilon_c \sim 25\%$
Test B: Rapid Parameter Variation (GAAC)

Sweep SSH parameters rapidly (non-adiabatic regime) while maintaining topological protection:

$$t_A(t) = t_A^{(i)} \left[1 + 0.1 \sin\left(\frac{2\pi t}{T} \right) \right] \tag{13}$$

Vary sweep timescale T from 100 ps to 1 ps. Prediction: - Winding number remains quantized even at 1 ps sweep rates - Validates your statement: "system breaks systematically when forced"

Test C: Parameter Deviation

Intentionally detune SSH parameters away from optimal τ_i values:

$$t_A \to t_A(1+\delta), \quad \delta \in \{-0.05, -0.02, 0, +0.02, +0.05\} \tag{14}$$

Prediction: - For $|\delta| < 0.01$: N_{wind} remains locked - For $|\delta| > 0.02$: N_{wind} jumps discontinuously (topological transition) - System exhibits no smooth variation—characteristic of topological protection

4 Expected Outcomes

4.1 Success Scenario

If measured winding numbers match $N_{\text{wind}} = \tau_i$:

- ✓ Confirms Metatime topological mechanism is physical, not mathematical artifact
- ✓ Demonstrates topological protection prevents arbitrary deformations
- ✓ Provides independent verification before DUNE neutrino results
- ✓ Elevates Metatime from theoretical prediction to experimentally validated framework
- ✓ Grade: 90/100 → 95/100

4.2 Partial Success

If winding numbers deviate by $\sim$ 5–10% but exhibit systematic pattern:

- $\sim$ Indicates underlying topological mechanism is correct
- $\sim$ Suggests refinement needed in τ_i eigenvalue computation
- $\sim$ Motivates Phase 2 analysis with refined parameters
- $\sim$ Grade: 85/100

4.3 Null Result

If winding numbers show no correlation with τ_i:

- × Falsifies SSH-Metatime mapping (but not Metatime itself)
- × Indicates topological eigenvalues cannot be tested optically
- × Redirects focus to DUNE neutrino experiments
- × Still valuable: establishes experimental limits
- × Grade: 75/100 (provides information)

5 Experimental Resources Required

5.1 Equipment

- Femtosecond laser writing system (1 kHz repetition, 50 fs pulses)
- 1550 nm pulsed pump laser (45 W peak power capability)
- Near-field scanning optical microscopy (NSOM) system
- Phase-shifting interferometer for wavefront measurement
- Spatial light modulator (SLM, 512×512 pixels, 1550 nm)

Estimated budget: \$250k–\$400k (mostly existing equipment in photonics labs)

5.2 Collaborators

- **Optical soliton experts:** Reference [1] (temporal solitons in SSH)
- **Topological lasing researchers:** Reference [2] (reconfigurable solid-state lasing)
- **Accelerated topological pumping:** Reference [3] (GAAC method validation)

These groups have published results within last 3 months and possess required infrastructure.

6 Timeline

Phase	Timeline	Deliverable
1 (Fabrication)	Jan 15 – Mar 15, 2026	SSH waveguide array
2 (Soliton pumping)	Mar 20 – May 20, 2026	Compression/localization data
3 (Winding measurement)	May 25 – Jul 25, 2026	Quantized winding numbers
4a (Disorder robustness)	Jul 30 – Aug 30, 2026	Anderson localization results
4b (Rapid variation)	Aug 30 – Sep 30, 2026	GAAC validation
4c (Parameter deviation)	Oct 1 – Oct 31, 2026	Topological transition mapping
Paper	Nov 1 – Dec 15, 2026	*Nature Physics* submission
Total duration: 11 months		

7 Significance

This experimental program offers:

1. **Direct test of topological mechanism**: Verifies that fermion spectrum emerges from topological constraints, not accident
2. **Timeline advantage over neutrino experiments**: SSH results by Q4 2026 vs. DUNE pre-results in 2027–2030
3. **Cross-domain validation**: Same topological principles should govern both optics and particle physics
4. **Publication strength**: Unique combination of theory + optical experiment + particle physics predictions
5. **Falsifiability**: Clear null-result scenarios; not protected against experiment

8 Conclusion

The Metatime topological eigenvalue hypothesis is amenable to direct experimental test using recent advances in photonic SSH lattices (Dec 2025 – Jan 2026). By mapping fermion mass eigenvalues to optical winding numbers and measuring soliton dynamics, we can validate the framework independent of DUNE neutrino measurements.

This program is **doable within 11 months** with existing experimental infrastructure and requires collaboration with recent topological photonics teams.

Recommended next step: Contact experimental groups at [1], [2] to discuss SSH lattice design and soliton measurement protocols.

References

[1] Observation of temporal optical solitons in topological SSH waveguides. *Nature Communications*, Nov 2024.

[2] Reconfigurable all-solid-state topological lasing at arbitrary sites. *Nature Photonics*, Jan 2026.

[3] Accelerated topological pumping in photonic waveguides. *Phys. Rev. Lett.*, 2023.

[4] A. Lipa, "Metatime: Time and Topological Derivation of Fermion Masses," 2025. Intention Lab, ResearchGate

[5] R. Penrose & S. Hameroff, "Consciousness in the Universe: Orch OR Revisited," *Physics of Life Reviews*, 2014.